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Research Paper

Effect of nanobubble water on medium chain carboxylic acids production in anaerobic digestion of cow manure

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ABSTRACT

Nanobubble water promotes the degradation of difficult-to-degrade organic matter, improves the activity of electron transfer systems during anaerobic digestion, and optimizes the composition of anaerobic microbial communities. Therefore, this study proposes the use of nanobubble water to improve the yield of medium chain carboxylic acids produced from cow manure by chain elongation. The experiment was divided into two stages: the first stage involved the acidification of cow manure to produce volatile acidic fatty acids as electron acceptors, and the second phase involved the addition of lactic acid as an electron donor for the chain elongation. Three experimental groups were established, and air, H_2, and N_2 nanobubble water were added in the second stage. Equal amounts of deionized water were added in the control group. The results showed that nanobubble water supplemented with air significantly increased the caproic acid concentration to 15.10 g/L, which was 55.03 % greater than that of the control group. The relative abundances of *Bacillus* and *Caproiciproducens*, which are involved in chain elongation, and *Syntrophomonas*, which is involved in electron transfer, increased. The unique ability of air nanobubble water supplemented to break down the cellulose matrix resulted in further decomposition of the recalcitrant material in cow manure. This effect subsequently increased the number of microorganisms associated with lignocellulose degradation, increasing carbohydrate metabolism and ATP-binding cassette transporter protein activity and enhancing fatty acid cycling pathways during chain elongation. Ultimately, this approach enabled the efficient production of medium chain carboxylic acids.

1. Introduction

The rapidly growing population has significantly increased milk and beef consumption, leading to the production of approximately 7.6 million tons of cow manure per year worldwide (Boldú and Pous, 2020; Jomnonkhaow et al., 2021). As biomass resources, organic waste such as cow manure can be converted into organic fertilizers, bioenergy and high value-added chemicals (Kunatsa and Xia, 2022). Cow manure is suitable as a substrate for anaerobic digestion (AD) due to its high moisture, high ash content, high lignocellulose content, abundance of alkali metals and AD microorganisms (Li et al., 2021). It has been confirmed that animal manure can generate medium chain carboxylic acids (MCCAs) through AD, with caproic acid being its primary product. Zhang et al. used swine manure as substrate and added ethanol as an electron donor for AD to produce MCCAs. As a result, the concentration of caproic acid reached 3.9 g/L (Zhang et al., 2020). Lian et al. found that the co-digestion of corn straw and cow manure to produce lactic acid at medium temperature simultaneously produced caproic acid, and the percentage of caproic acid could be 7.19–14.06 % (Lian et al., 2022). MCCAs are carboxylic acid molecules with 6–12 carbon atoms. It is a significant organic molecule that can be employed as an antibacterial agent, food additive, precursor of liquid biofuels, and growth-promoting antibiotic alternative in animal feed (Contreras-Dávila et al., 2021). Therefore, converting organic matter such as animal manure into MCCAs through AD can not only process waste and protect the environment, but also have high economic returns.

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The chain elongation (CE) reaction is the main process used to produce MCCAs by AD. The process uses ethanol or lactic acid as the electron donor and volatile fatty acids (VFAs) as the electron acceptor, and an electron donor is used to increase the length of the carbon chain of the electron acceptor to achieve the production of MCCAs (Wu et al., 2019a). Therefore, electron donors and acceptors are indispensable prerequisites for MCCAs production in AD. The Gibbs free energy values for the CE reaction using lactic acid and ethanol as an electron donors were −123.1 kJ/mol and −79.0 kJ/mol, respectively, and lactic acid has thermodynamic advantages over ethanol when used as an electron donor (Zhang et al., 2022b). By introducing lactic acid bacteria into waste that is high in carbohydrates, lactic acid can be generated, which makes the process more advantageous both ecologically and economically (Kang et al., 2022; Zhang et al., 2022b). For these reasons, CE reactions driven by lactic acid as an electron donor have received much attention in recent years. However, inefficiencies arise in the synthesis of MCCAs due to challenges such as low electron transfer efficiency, restricted substrate utilization by CE microorganisms, and competitive metabolism (Wu et al., 2019a; Wu et al., 2021b). If the rate-limiting process mentioned above can be increased during AD, MCCAs yield will be further improved.

Nanobubble is defined as bubbles less than 1000 nm in diameter. It has unique physical and chemical properties, such as a high negative Zeta potential, stability, and solubility, and generates reactive oxygen species (Chuenchart et al., 2021). Nanobubbles have a high-curvature gas-liquid interface, which has a very high surface energy and can therefore adsorb and aggregate microbial cells (Zhang et al., 2022c). Nanobubble water (NBW) has been investigated in AD biogas production and has shown good performance in promoting the degradation of lignocellulose, increasing the activity of the electron transport system (ETS), and optimizing the microbial community structure (Wang et al., 2020a; Wang et al., 2020b). For example, wang et al. investigated the effect of O_2-containing gas NBW on cellulose AD and showed that the addition of NBW reduced cellulose content and cellulose crystallinity by 8–14 % and 9–21 %, respectively, and ETS activity by 1.7-fold (Wang et al., 2020b). For example, Wang et al. reported that CO-NBW increased the methane production rate by 21.4 %, improved substrate degradation, enriched methanogenic bacteria, and optimized the composition of the microbial community (Wang et al., 2023b). The CE process depends on CE bacteria. CE bacteria are also anaerobic bacteria that coexist and cooperate with hydrolyzing and acidifying bacteria (Wang et al., 2023e). Hence, it will be intriguing to determine whether NBW enhance the degradation of recalcitrant organic matter, improve electron transfer efficiency, and optimize the microbial community structure in MCCAs producing systems. To our knowledge, this is the first time that the effect of NBW on MCCAs production performance has been evaluated.

Therefore, this research aimed to elucidate the effect of different types of NBW (air, H_2 and N_2) on the production of MCCAs from cow manure AD. The AD experiment comprised two stages: 1) The first stage was an acidification experiment where electron acceptor VFAs were produced from cow manure AD. 2) The second stage was a CE experiment where an exogenous electron donor (lactic acid) was added to the first-stage products to produce MCCAs. NBW was added in the second stage to assess the effect of NBW on MCCAs synthesis. Then, the effects of the properties of NBW, such as negative Zeta potential, release of reactive oxygen species, high stability, and high mass transfer efficiency, on substrate degradation, intermediate product production, electron transfer efficiency, key enzyme activities and anaerobic microbial community structure in the AD process were investigated to explore the mechanism of action of NBW. This research contributes to the development of sustainable solutions to increase the production of high value-added products from biomass resources.

2. Materials and methods

2.1. Substrates and inoculums

Cow manure was obtained from Zhongxin Dairy Farming Professional Cooperative, Xinghua City, Jiangsu Province, China. The samples were stored at −20 °C. The inoculum for acidification was collected from the Taihu New City Wastewater Treatment Plant, Wuxi City, Jiangsu Province, China. The samples were subjected to heat shock treatment at 75 °C for 3 h to inhibit the activity of methanogenic bacteria (Arslan et al., 2016). The inoculum for CE was collected from the running CE reactor in the laboratory. During reactor operation, cow manure is employed as the substrate, exogenous lactic acid serves as the electron donor, and caproic acid is the primary product. The physical and chemical indicators of the inoculum and cow manure are shown in Table S1.

2.2. Preparation of NBW

NBW was prepared using a nanobubble generator (Shanghai Zhongjing Environmental Protection Technology Co., Ltd., ZJC-NM, China), where deionized water and gas (air, H_2, and N_2) were fed into (150 mL/min) the nanobubble generator (Fig. 1).

2.3. AD experimental setup

2.3.1. Acidification of cow manure

First, 143 g of cow manure, 30 g of inoculum used for acidification, and 277 g of deionized water were added to a 500-mL glass serum bottle and mixed thoroughly. The pH in the glass serum bottle was adjusted to 6.5 ± 0.1 with 5 M HCl and NaOH, after which the glass serum bottle was sealed. The cap of the bottle had two holes to connect the silicone tubes for sampling during fermentation. After sealing, the headspace portion of the glass serum bottle was purged with pure nitrogen for 10 min to expel the air inside the bottle and create an anaerobic environment.

2.3.2. CE experimental setup

When lactic acid acts as an electron donor, the CE process includes two metabolic pathways, including reverse β oxidation (RBO) and fatty

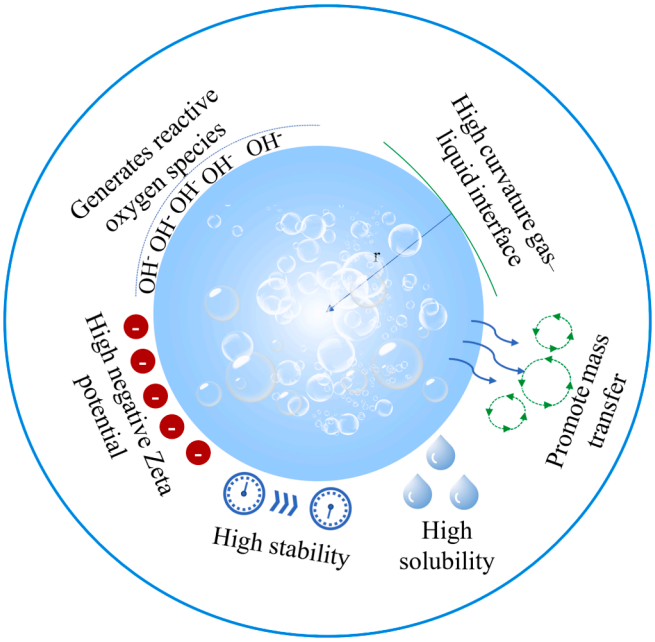


Fig. 1. The unique properties of the NBW.

acid cycling (FAB), as shown in Fig. 2. The fermentation substrate for the CE experiment was the digest produced in Section 2.3.1. To ensure that the fermentation substrate was the same for the different treatment groups, all the acidified fermentation products generated in Section 2.3.1 were mixed together and stirred well. Then, 250 mL of the mixture was taken and added to 500-mL glass serum bottles for the CE experiments. The inoculum for the CE experiments was obtained from the running CE reactor, and 50 mL was added to each glass serum bottle. Then, 100 mL of NBW or deionized water. The three experimental groups were Air-NBW, H_2-NBW and N_2-NBW, which were named Air, H_2 and N_2, respectively, and deionized water served as the control group, which was named CK. Three replicates were used for each experimental group. The lactic acid/VFAs ratio was set to 3. VFAs were derived from the fermentation of cow manure produced during the acidification phase, and the initial VFAs concentrations in the glass serum bottles was 9.83 ± 0.21 g/L. Therefore, 11.8 g of lactate was added to each glass serum vial, resulting in a final lactate concentration of 29.5 ± 0.12 g/L. The pH inside the glass serum vials was then adjusted to 5.5 ± 0.1 with 5 M NaOH. The caps were tightly screwed on, and the tops of the glass serum vials were emptied and purged with nitrogen gas for 10 min to create an anaerobic environment. The glass serum vials for the CE experiments were placed in a water bath (30 ± 1 °C) for 18 days (see Table 1). The selection of these three gases for NBW preparation is based on the following considerations. Air is cost effective and readily available without the need for purchase, and previous studies have indicated that a microaerobic environment can enhance AD (Cao et al., 2022). Exogenous supplementation of hydrogen gas improves electron transfer efficiency and increase MCCAs production (Wu et al., 2019b); thus, it is important to investigate the effects of NBW prepared with hydrogen gas. Nitrogen gas is commonly used to create anaerobic environments, and nitrogen gas NBW contributes to the enhancement of AD performance and lignocellulose degradation (Yang et al., 2020).

Liquid samples were collected every two days for the determination of pH, VFAs, lactic acid, ammonia nitrogen, and MCCAs concentrations. After AD, the degasification mixture samples were subjected to extracellular polymeric substance (EPS), LDH, oxidation reduction potential (ORP), cyclic voltammetry (CV), ETS, electron accept capacity (EAC), electron doner capacity (EDC), electrochemical impedance spectroscopy (EIS), I–t, and microbial community composition analyses. The biogas residue was freeze–vacuum dried, and substrate degradation was characterized using X-ray diffraction (XRD), X-ray photoelectron spectroscopy (XPS), and Fourier transform infrared spectroscopy (FTIR). The biogas slurry was filtered through a 0.45-μm filter membrane, and the filtered solution was dissolved organic matter (DOM). XRD, FTIR and EEM were used to characterize the DOM to analyze whether more substrates were converted into DOM.

2.4. Analytical methods

For the detection of MCCAs using gas chromatography (GC-2030, Shimadzu), a gas chromatograph was used. The temperatures of the injector and detector were 275 and 300 °C, respectively, and the capillary column temperature was increased from 70 to 170 °C at 20 °C/min and increased to 240 °C at 20 °C/min after 5 min. Lactic acid was detected using high-performance liquid chromatography (LC-1260, Agilent) on an HSS T3 column (4.61 × 150 mm, 5 μm, Waters) with a mobile phase of 20 mM sodium dihydrogen phosphate (pH 2.7), an injection volume of 10 μL, a detection wavelength of 210 nm, and a column temperature of 30 °C.

Total solids (TS), volatile solids (VS), pH, CV, EIS, ORP, XRD, total carbon and total nitrogen content, ammonia nitrogen concentration, and VFAs concentration were determined as reported in our previously published article (Liu et al., 2022). The I–t curve was measured at constant voltage for 5 min at +0.6 V (positive potential) and −0.6 V (negative potential) (He et al., 2014). The methods used for EAC and EDC detection are described in the literature (He et al., 2014). The different fractions of EPS, such as loose-bound EPS (LBEPS) and tight-bound EPS (TBEPS), were extracted following previously reported methods (Wu et al., 2021c; Yang et al., 2019). Lactate dehydrogenase (LDH) was measured using an LDH kit (Solarbio, China). ETS activity was determined using the INT assay (Hou et al., 2022). Microbial community analysis was conducted using high-throughput sequencing of 16S rRNA, and the PCR amplification primer 515F-806R was selected. GENE DENOVO was commissioned to perform high-throughput

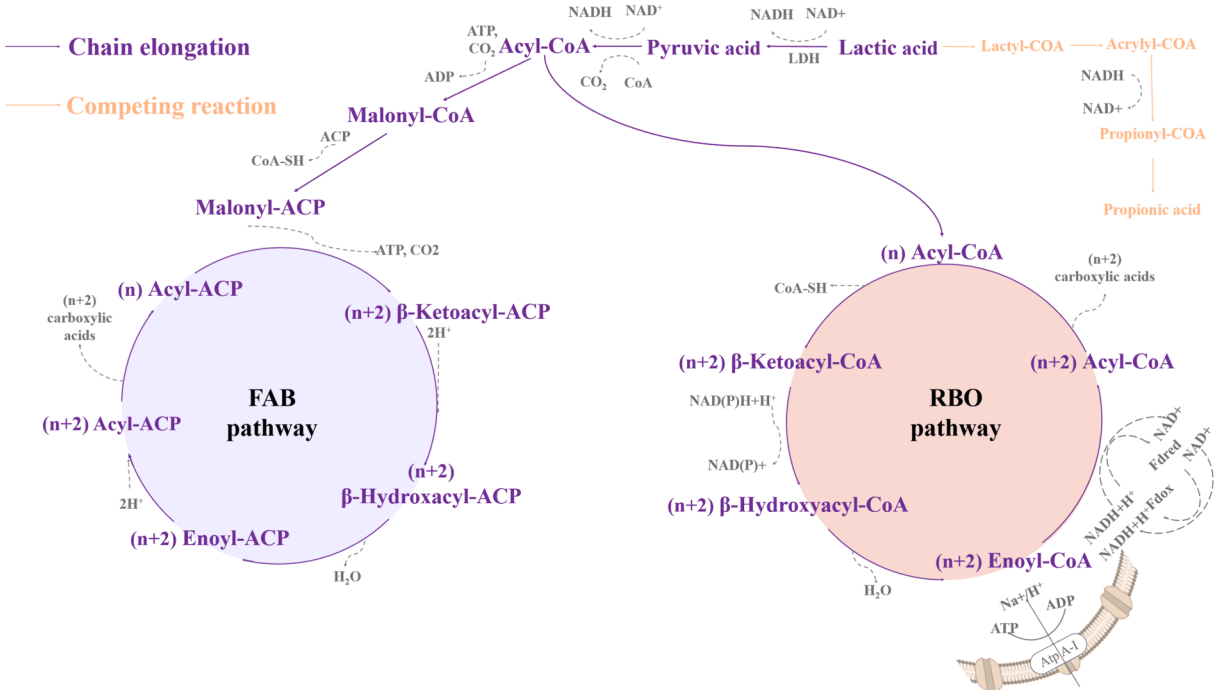


Fig. 2. Metabolic pathways for MCCA production in AD.
Adapted from (Wu et al., 2023).

Table 1
Physical and chemical properties of at the beginning of the CE experiment.

Name	VS/%	TS/%	TCOD/(mg/L)	SCOD/(mg/L)	VFAs/(g/L)	Lactic acid/(g/L)	Time/d	pH	Temperature/°C	Typology of NBW
CK	79.41 ± 1.79	4.70 ± 0.11	5589.33 ± 7.54	3338.67 ± 8.56	9.83 ± 0.21	29.5 ± 0.12	18	5.5 ± 0.1	30 ± 1	–
Air	79.41 ± 1.79	4.70 ± 0.11	5589.33 ± 7.54	3338.67 ± 8.56	9.83 ± 0.21	29.5 ± 0.12	18	5.5 ± 0.1	30 ± 1	Air
H_2	79.41 ± 1.79	4.70 ± 0.11	5589.33 ± 7.54	3338.67 ± 8.56	9.83 ± 0.21	29.5 ± 0.12	18	5.5 ± 0.1	30 ± 1	H_2
N_2	79.41 ± 1.79	4.70 ± 0.11	5589.33 ± 7.54	3338.67 ± 8.56	9.83 ± 0.21	29.5 ± 0.12	18	5.5 ± 0.1	30 ± 1	N_2

sequencing analysis and community composition analysis using its cloud analysis platform (https://www.omicsmart.com/). Three parallel experiments were performed.

XPS was performed with a Thermo Scientific K-Alpha+, and EEM was performed with a Hitachi F-7000. A Rigaku SmartLab SE was used for XRD, and a Thermo Scientific Nicolet iS20 was used for FTIR. The XPS test track was C1s. The EEM emission and excitation wavelengths were 220–500 nm. The excitation wavelength interval was 10 nm, and the emission wavelength interval was 5 nm. The scanning speed was 2400 nm/min, and the excitation/emission slit width was 5 nm. The XRD scanning angle was 5–80°, and the scanning speed was 10°/min. The FTIR test wavelength ranged from 400 to 4000 cm^{-1}. The Zeta potential was tested using a nanoparticle size potentiometer (NICOMP Z3000, USA).

2.5. Data analysis

For analysis, the experimental data were fed into SPSS 26.0 (Microsoft, USA), while OriginPro 2023b (Origin Lab, USA) was used to generate figures. During the CE experiment, liquid samples were collected from the experimental facility every two days to detect the concentrations of MCCAs and VFAs. In order to ensure the reliability and accuracy of the experimental results, we prepared three glass serum vials for each experimental group as repeat samples, so three sets of data will be obtained for each sample. Subsequently, we imported these data into SPSS 26.0 statistical software for in-depth analysis. By performing the single-factor ANOVA test, we were able to calculate the p-value of the difference significance between the different groups to determine whether there was a statistically significant difference between the groups. In order to understand more precisely which experimental groups have significant differences, we further applied the Tukey HSD

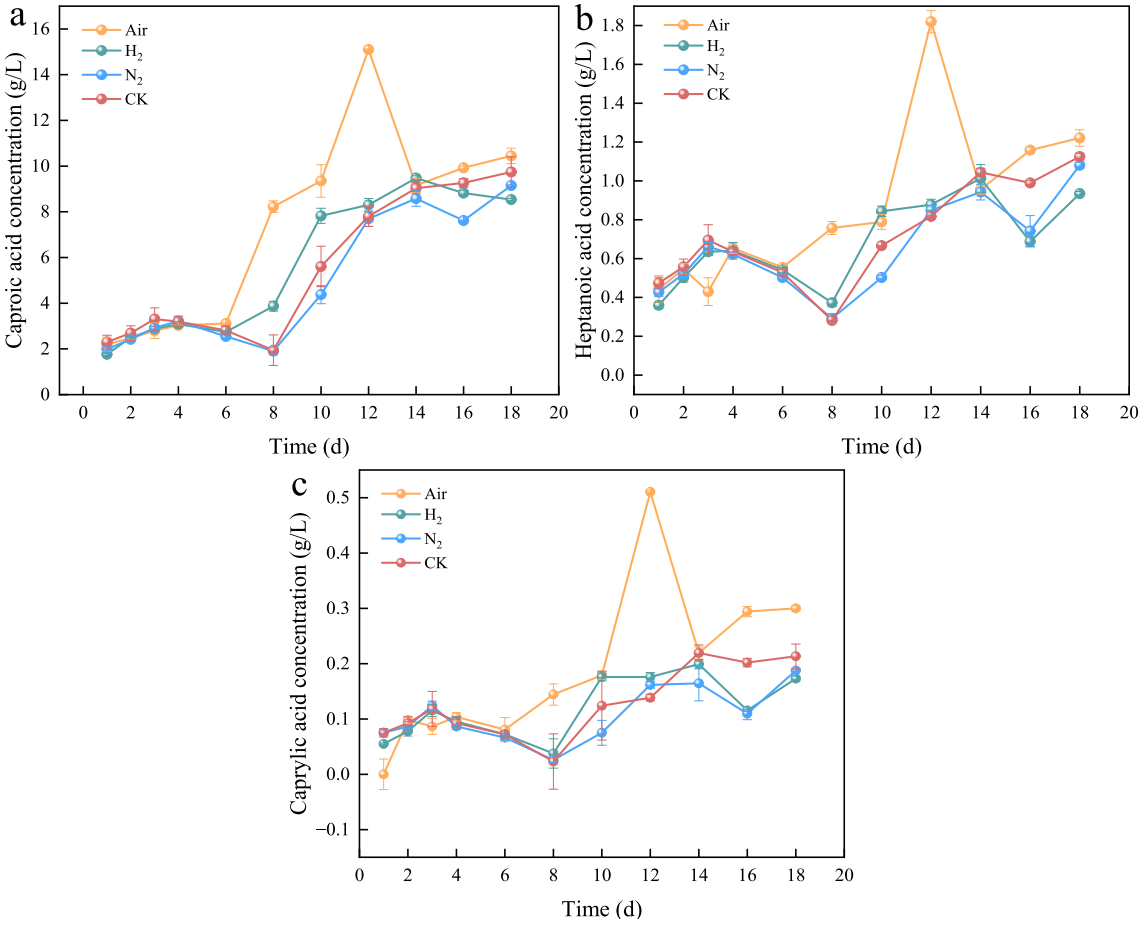


Fig. 3. Caproic acid (a), heptanoic acid (b) and caprylic acid (c) concentrations in groups treated with different nanobubble water. All data were the averages from three parallel experiments.

multiple comparison method.

3. Results and discussion

3.1. Impact of nanobubble water on MCCAs production from cow manure

To investigate the influence of various types of NBW on the production of MCCAs, distinct experimental groups were established using air, H_2, and N_2. As shown in Fig. 3, significant variance in the concentrations of MCCAs (specifically caproic acid, heptanoic acid, and caprylic acid) were noted among these three types of NBW. For the initial 6 days of AD, the MCCAs concentrations in each experimental group exhibited minor fluctuations between 2 and 4 g/L, devoid of obvious alterations. However, the MCCAs concentrations of the Air experimental groups increased notably on the 8th day of the experiment, peaking on the 12th day. The peak concentrations of caproic acid, heptanoic acid, and caprylic acid were 15.10, 1.82, and 0.51 g/L, respectively. The peak concentrations of caproic acid, heptanoic acid, and caprylic acid in the Air experimental group were 55.03, 62.50, and 131.82 % greater than those in the CK experimental group (9.74, 1.12, and 0.22 g/L, respectively). Tukey HSD analysis revealed that considering 95 % confidence intervals, the MCCAs concentrations in the Air experimental group were significantly greater than those in the CK experimental group ($p < 0.05$). An increase in the concentrations of MCCAs of more than 50 % indicates that air-NBW has a significant effect on MCCAs production. The H_2 experimental group reached peak concentrations of 9.47, 1.01, and 0.20 g/L for caproic acid, heptanoic acid, and caprylic acid, respectively. 9.15, 1.08, and 0.19 g/L for caproic, heptanoic, and caprylic acids, respectively, in the N_2 experimental group. The concentration of caproic acid in the H_2 experimental group did not significantly differ from that in the CK experimental group ($p = 0.358$). Moreover, the concentrations of heptanoic acid and caprylic acid were significantly lower than those in the CK group ($p < 0.05$). The concentration of caproic acid was significantly lower in the N_2 experimental group than in the CK experimental group ($p < 0.05$). The concentration of heptanoic acid did not significantly differ from that in the CK experimental group ($p > 0.05$). The concentration of caprylic acid was also significantly lower than that in the CK experimental group ($p < 0.05$). The main component of the MCCAs in each experimental group was caproic acid, and air-NBW had a significant promoting effect on caproic acid.

Different types of NBW exhibited different effects in comparison to those in the CK experimental group. The concentrations of MCCAs in the Air experimental group increased most significantly from 6 to 12 days. The variation in the impact of NBW type on MCCAs synthesis was potentially attributed to the gas type of the NBW species since the same preparation process for NBW was employed in this study, except for the difference in gas type. The concentration of caproic acid used in this study is notable. For instance, Zhang et al. reported that the highest concentration of caproic acid in their investigation of MCCAs production from swine manure AD was 3.9 g/L (Zhang et al., 2020). The concentration of caproic acid reached a high level (15.10 g/L) in this study compared with previous studies. Wang et al. investigated the impact of nanobubble water on AD and found that nanobubbles enhance microbial activity by promoting ATP synthesis. Moreover, the use of oxygenated nanobubble water led to a boost in the electron transfer efficiency of the AD system (Wang et al., 2020b). ETS activity in the AD system reflects the respiratory activity of anaerobic microorganisms (Wang et al., 2022). Deng et al. showed a significant positive correlation between caproic acid yield and ETS activity (Deng et al., 2024). Therefore, the highest levels of MCCAs in the Air experimental group may be related to microbial activity, which specifically needs to be analyzed in terms of microbial community structure.

3.2. Effect of nanobubble water on key physicochemical factors

Fig. S1a-B shows that the bubble sizes in the three NBWs are mainly between 100 and 600 nm. The bubble concentration in different NBWs at 200 nm is the highest compared to other sizes. The total nanobubble concentration was the highest for Air-NBW followed by N_2-NBW, and the lowest concentration was noted for H_2-NBW. Fig. S1c-e shows the nanobubble Brownian motion images, demonstrating that the water samples with different NBWs contained nanobubbles. NTA revealed the successful preparation of different types of NBW with different concentrations and sizes. The Zeta potential of Air-NBW was -13.83 ± 0.26 mV, that of H_2-NBW was -9.03 ± 0.63 mV, and that of N_2-NBW was -15.13 ± 0.30 mV. Significant differences in the Zeta potentials of the three types of NBWs were noted ($p < 0.05$). The Air experimental group had the highest concentration of smaller nanobubbles and the highest concentration of caproic acid. The change in the concentration of caproic acid may be related to these properties. VFAs are the electron acceptors of the CE process and undergo carbon chain lengthening by reacting with acetyl coenzyme A generated by the oxidation of electron donors, yielding two carbon atoms each cycle (Wu et al., 2021a). Thus, the CE process entails the conversion of acetic acid to butyric acid, butyric acid to caproic acid, caproic acid to octanoic acid (even carbon), and propionic acid to valeric acid, followed by the conversion of valeric acid to heptanoic acid (odd carbon). As shown in Fig. 4 a–d, on the 12th day of the experiment, the butyric acid concentration in the Air experimental group was 2.03 g/L, which was significantly lower than the 4.00 g/L in the CK experimental group ($p < 0.05$). The valeric acid concentration in the Air experimental group was 1.58 g/L, which was significantly lower than the 2.60 g/L in the CK experimental group ($p < 0.05$). This change indicated that the CE reaction could quickly synthesize MCCAs in the Air experimental group. In addition, butyric acid and valerate concentrations were lower in the air group compared with the other groups after the CE reaction began, indicating that the CE reaction rate was faster. In addition, the precursors (butyric acid and valerate acid) to the synthesis of MCCAs were rapidly consumed, so the concentrations were lower. We also observed that the propionic acid concentration in the air experimental group was lower than that in the other groups. When lactic acid serves as the electron donor, propionic acid is produced as a byproduct through the propionate pathway (Fig. 1), forming a competitive relationship with the MCCAs metabolic pathway (Ma et al., 2022). The addition of Air-NBW may influence the microbial community structure, weakening the competitive pathway of propionic acid, and allowing more lactic acid to be utilized for MCCAs production.

Lactic acid serves as an electron donor for the CE reaction, so the consumption of lactic acid reflects CE reaction progression. As shown in Fig. 4e, lactic acid depleted faster in the air experimental group than in the CK experimental group, with a concentration of only 1.07 g/L by Day 8. By Day 10, the lactic acid level was 0.05 g/L, indicating that the addition of Air-NBW led to earlier consumption of the electron donor and accelerated the carboxylate platform fermentation process. The experimental group exposed to H_2 showed a reduction in lactic acid to 17.52 g/L on Day 8, which continued to decrease to 0.03 g/L on Day 10. The lactic acid concentration changes in the N_2 and CK experimental groups were similar; they began to decrease on the 10th day and reached 0 g/L by the 12th day. The Air experimental group displayed a significant increase in the MCCAs concentrations, indicating that Air-NBW accelerated the conversion of lactic acid to acetyl coenzyme A, thereby accelerating the CE reaction. The increase in the concentrations of MCCAs is potentially related to the improvement in electron transport efficiency and the optimization of the microbial community structure in the AD system.

In the MCCAs production process, pH is a key index used to represent the stability of the AD system. The pH changes exhibited by each experimental group during AD are shown in Fig. 4f. During the initial 6 days of fermentation, the pH remained close to the initial value of 5.5. Compared with that on the first day, the pH of each experimental group on Day 6 did not significantly change, and the p value ranged from 0.496 to 1. The pH of the air treatment group increased to 7.16 on the 8th day,

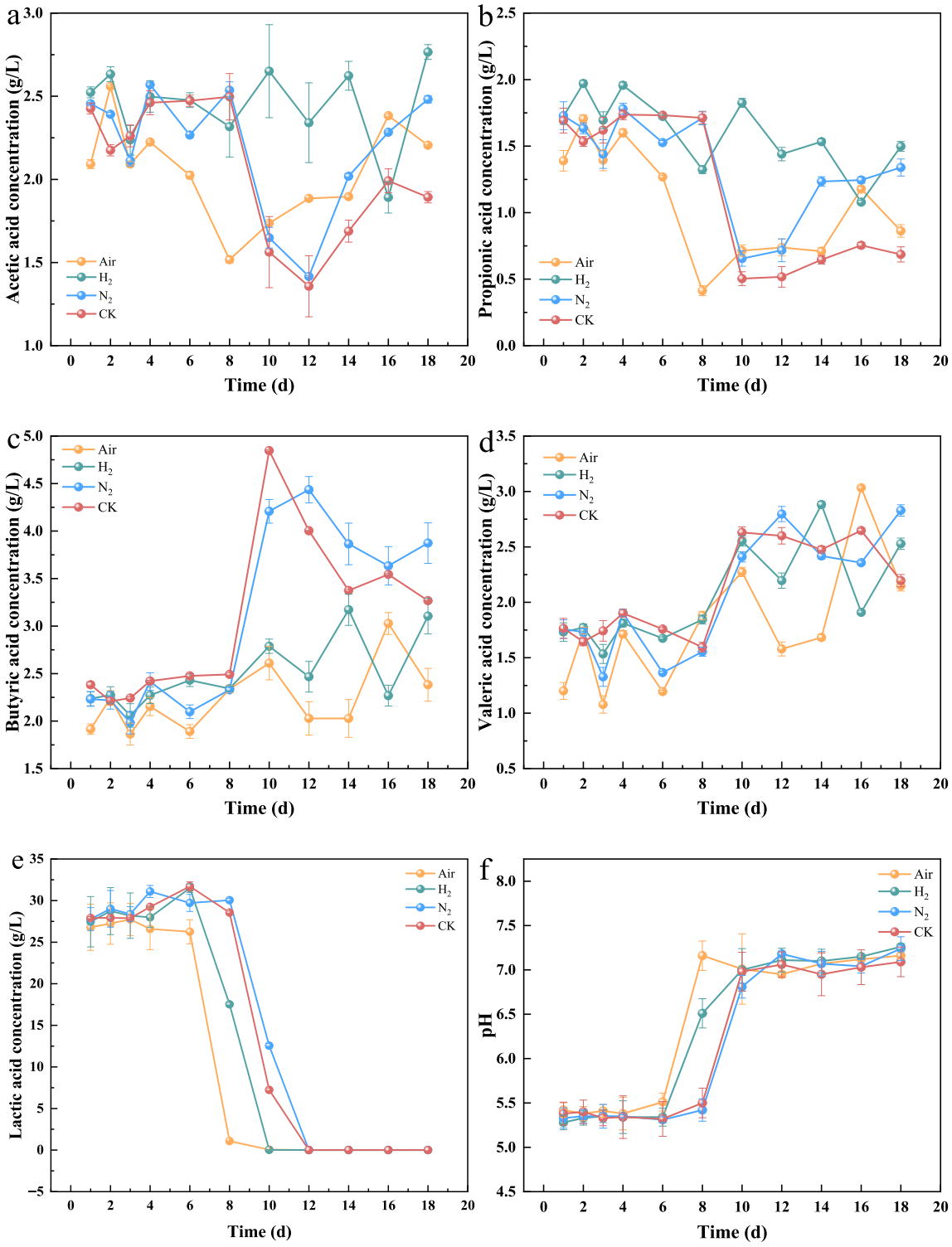


Fig. 4. VFAs concentration (a-d), lactic acid concentration (e), pH (f), ammonia nitrogen concentration (g), LDH concentration (h), Protein (i), and Polysaccharide concentrations (j) of EPS in groups treated with different nanobubble water. All data were the averages from three parallel experiments.

and this value was maintained. However, the N_2 and CK experimental groups exhibited growth on Day 10 and began to stabilize at a pH of approximately 7 on Day 12. In lactate-driven CE reactions, the pH varies in the range of 5–7. Montecchio et al. reported that the most suitable pH for lactate elongation was 6 (Montecchio et al., 2024). Our results showed some discrepancies, but the optimal pH range was close to the reported optimal pH range. This observation may be attributed to the enrichment of CE microorganisms, altering the chemical environment of the AD system during the CE process.

The ammonia nitrogen concentration has an effect on CE bacteria, and a high ammonia nitrogen concentration (2.1 g/L) inhibits MCCAs synthesis (Zhang et al., 2019). The ammonia nitrogen concentration during the AD process of each experimental group is illustrated in Fig. 4g. The Air experimental group exhibited ammonia nitrogen

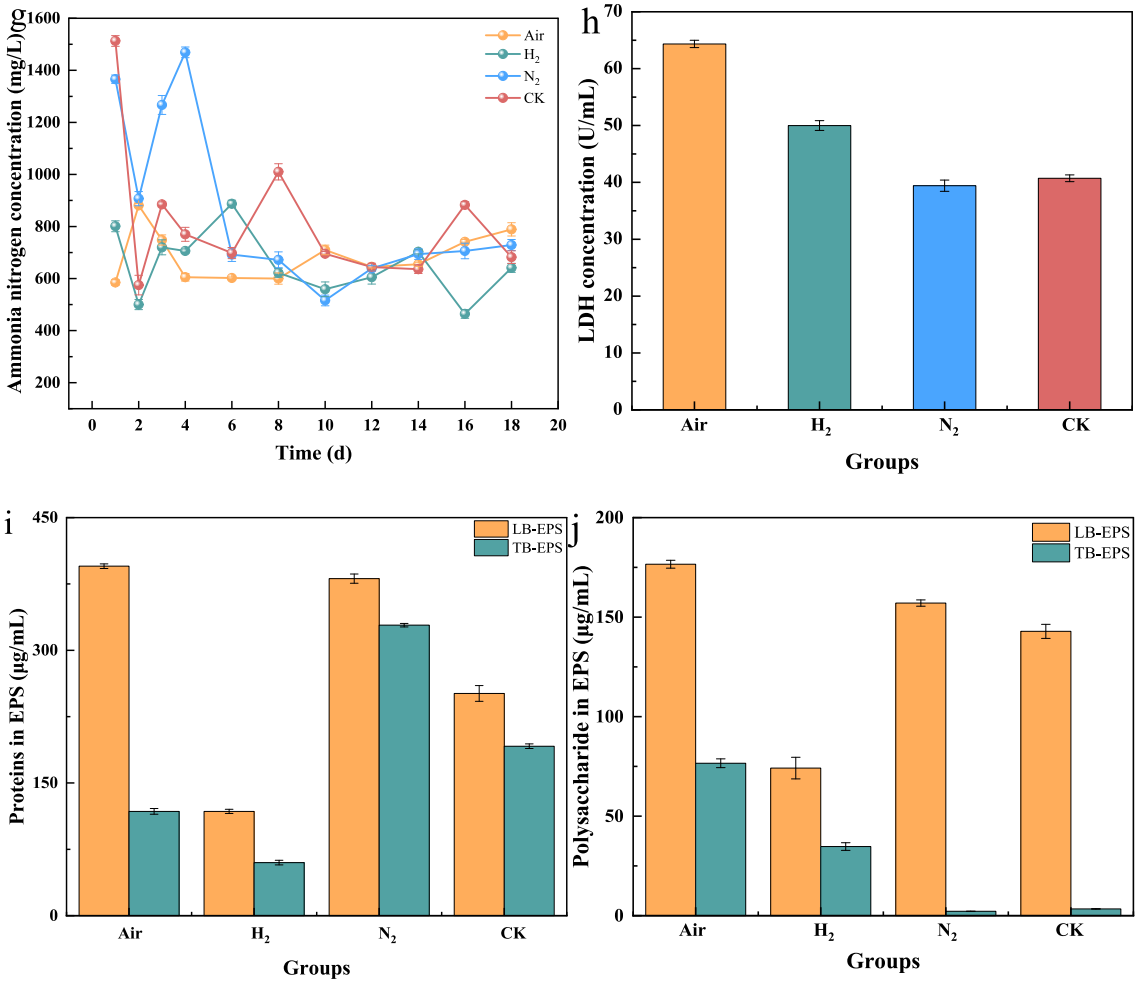


Fig. 4. (*continued*).

concentration ranging from 600 to 800 mg/L, whereas the H_2 treatment group exhibited ammonia nitrogen concentration ranging from 500 to 850 mg/L. The N_2 and CK experimental groups showed peak nitrogen levels of 1469 and 1513 mg/L, respectively, taking ten days to achieve relative stability. The N_2 and CK treatments had adverse effects on the CE process and outcome. The lower ammonia concentration in the Air experimental group suggests that Air-NBW also has the effect of reducing ammonia nitrogen in the AD system for the production of MCCAs, maintaining the CE bacteria in a suitable ammonia nitrogen environment. The decrease in ammonia nitrogen concentration observed with Air-NBW is potentially associated with the heightened activity of anaerobic bacteria. During the production of MCCAs, anaerobic bacteria utilize ammonia nitrogen as a nitrogen source for cellular activities (Jiang et al., 2019). The highest MCCAs yield was correlated with increased anaerobic bacterial activity, consequently leading to greater consumption of ammonia nitrogen.

Lactate-based CE relies on LDH to oxidize lactate to pyruvate and further react with acetyl coenzyme A (Kim et al., 2022); thus, LDH can respond to the effect of different NBWs on lactate oxidation. The LDH concentrations are shown in Fig. 4h. The LDH concentration of the Air experimental group was 64.35 U/mL, which was 58.03 % greater than that of the CK experimental group (40.72 U/mL). The concentration in the H_2 experimental group was 50.00 U/mL, which was 22.79 % greater than that in the CK experimental group. At 39.41 U/mL, the concentration in the N_2 experimental group was slightly lower than that in the CK experimental group. The LDH concentration among the experimental groups and the peak MCCAs concentrations exhibited high consistency. This finding implies that Air-NBW increased LDH activity in AD, hastened the oxidation of the electron donor lactate, accelerated the production of acetyl coenzyme A, and accelerated the CE reaction.

EPS can serve as an energy source for microorganisms, and EPS proteins possess electrochemically active compounds that facilitate electron transfer between bacteria (Song et al., 2017). As shown in Fig. 4i-j, EPS consisted predominantly of proteins and polysaccharides, with protein being the key component, indicating that Air-NBW notably stimulated proteins secretion in EPS. The Air experimental group exhibited the highest levels of protein and polysaccharides, with protein mainly accumulating in LB-EPS at a concentration of 395.09 μg/mL, exhibiting a 57.26 % increase from that of the CK experiment group (251.23 μg/mL). The LB-EPS of the Air experimental group had a polysaccharide concentration of 176.59 μg/mL, indicating a 23.85 % increase compared to that of the CK experimental group, which had a concentration of 142.85 μg/mL. Wang et al. reported that the addition of iron oxide increased the secretion of electroactive EPS, stimulated ethanol oxidation and facilitated electron transfer by CE bacteria (Wang et al., 2023c). In t this study, an increase in proteins and polysaccharides were incorporated into EPS. Protein was identified as the main component of EPS, which could create a conducive microbial environment for lactate oxidation and CE.

The ORP in the AD system impacts microorganism growth in the reactor. A lower ORP indicates more strict anaerobic conditions (Xu et al., 2014). The ORP in the NBW experimental group was greater than that in the CK experimental group (Fig. S2), suggesting that the addition of NBW affected the anaerobic environment. The ORP of the Air experimental group was the highest and was significantly greater than that of the CK experimental group ($p < 0.05$). However, no significant

correlation was detected between the ORP and MCCAs concentrations.

3.3. Understanding the impacts of nanobubble water on the electron transfer efficiency

CE is primarily facilitated through electron transfer between the electron donor and electron acceptor for completion of the redox reaction. A high electron transfer efficiency accelerates the CE reaction; thus, the electron transfer efficiency of the AD system was examined to assess the CE efficiency. Fig. 5a presents the CV curves from all experiments, demonstrating that the inclusion of NBW in the experimental group resulted in reduction peaks. Notably, the Air experimental group displayed the largest reduction peak current, followed by the H_2 experimental group and the N_2 experimental group, and the smallest reduction peak current was observed in the CK experimental group. The CV curves show exhibit the extracellular electron transfer capacity and the correlation between electroactive microorganisms in the AD system (Feng et al., 2018). A comparison of the CV curve outcomes indicates that the incorporation of NBW enhances the electrochemical activity of the AD system.

The EIS curves for each experimental group are shown in Fig. 5b. The slope of the curve in the low-frequency region of the Nyquist curve provides insight into the resistance of the electrolyte. The higher the slope is, the greater the resistance (Choi et al., 2017). The slope of the CK experimental group was the greatest, followed by that of the N_2 experimental group. The slope of the H_2 experimental group was reduced compared with that of the N_2 experimental group but greater than that of the Air experimental group. The slope of the Air experimental group was the smallest, resulting in a lower electron transfer resistance, which is beneficial for enhancing the electron transfer efficiency.

Fig. S3 displays the I-t curves of various experimental groups at a constant voltage. The EDC and EAC are calculated by integrating the current curves under applied constant positive and negative voltages. The EAC and EDC were consequently calculated by integrating the current curves (Fig. 5c). The EAC and EDC of the Air experimental group were 178.55 and 30.66 μmol/g C, respectively. In contrast, the EAC and EDC of the CK experimental group were 119.11 and 113.46 μmol/g C, respectively. The addition of Air-NBW significantly enhanced the EAC ($p < 0.05$) while reducing the EDC ($p < 0.05$). The H_2 and N_2 experimental groups displayed similar increases in the EAC and decreases in the EDC, suggesting that the addition of NBW is favourable for enhancing the reduction potential of the AD system. This study also examined, for the first time, the effect of NBW on the EAC and EDC of the MCCAs system used to produce AD.

After the addition of NBW, the ETS activity in all three experimental groups surpassed that in the CK group (Fig. 5d). The Air experimental group demonstrated the highest ETS activity (98.88 mg/g-h), which exceeded that of the CK experimental group by 40.75 %. The ETS activities of the H_2 experimental group (82.36 mg/g-h) and the N_2 experimental group (80.25 mg/g-h) exceeded those of the CK experimental group by 17.24 % and 14.23 %, respectively. The inclusion of NBW considerably enhanced the ETS activity of the AD system, where Air-NBW exhibited the most favorable impact. ETS activity characterizes the electron transfer activity of AD systems and microbial activity (Yin

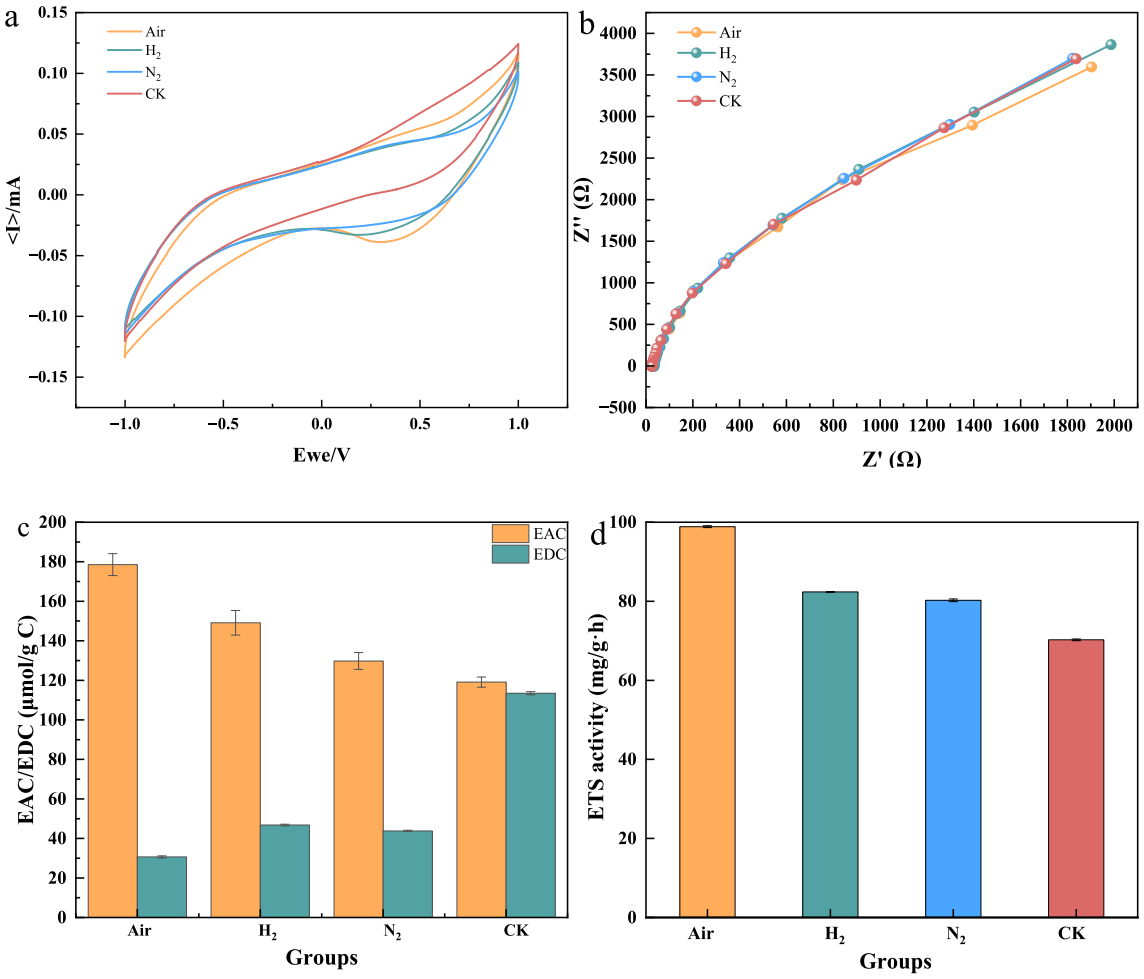


Fig. 5. Cyclic voltammetry characteristic curve (a), Nyquist curves (b), EAC/EDC (c), and ETS concentration (d) in groups treated with different nanobubble water. All data were the averages from three parallel experiments.

et al., 2018). The enhanced ETS activity suggested that Air-NBW stimulates electron transfer between microorganisms in the lactate-driven AD system.

3.4. Understanding the impact of NBW on cow manure degradation

The AD efficiency of cow manure is limited by the presence of components such as lignin, which are difficult to decompose. Evaluating the degradation level of the fermentation substrate, namely, cow manure, plays a crucial role in obtaining a better understanding of this mechanism. XPS was employed to characterize the chemical composition of the carbon in the digestate at the end of AD (Fig. 6a). Here, the C1s spectrum is separated into three peaks: peak 1 represents C-C, which is primarily derived from hydrocarbon organics such as polysaccharides; peak 2 embodies C-O and C-N; and peak 3 corresponds to COOR (Yin et al., 2015). Among all the experimental groups, peak 1 had the highest percentage, with 54.13 % in the Air experimental group, 61.4 % in the H_2 experimental group, 62.27 % in the N_2 experimental group, and 79.42 % in the CK experimental group. The addition of NBW noticeably decreased the area share of peak 1. Yu et al. analyzed the rupture of surface C-C resulting in the conversion of lignin from solid to liquid (Yu et al., 2022). That is, NBW assists in the breakdown of lignin in cow manure.

The EEM spectra (Fig. 6b) can indicate alterations in organic compounds within DOM (Song et al., 2023). The inclusion of Air-NBW resulted in a substantial increase in the humic acid levels within the DOM. This points towards increased degradation of organic matter present in the substrate. As it contains numerous oxygen-containing functional groups that transfer electrons within the AD system, humic acid plays a critical role in promoting the breakdown of complex substrates in anaerobic environments (Zhao et al., 2023). The higher humic acid content of the Air experimental group corresponded to its highest MCCAs concentrations and electron transfer efficiency.

XRD analysis was conducted on both the DOM and digestate (Fig. 6c). The diffraction peaks consistently appeared at 2-θ in all the experimental groups, indicating similar crystal structures. The DOM and air in the digestate experimental groups exhibited the lowest crystallinity, whereas those in the CK experimental group had the highest crystallinity. The low degree of crystallinity indicates that the structural integrity of the substance's crystals has been compromised (Olatunji and Madyira, 2023). Furthermore, the introduction of NBW amplifies the disturbance of the crystal structure of the organic matter.

The differences in the functional groups of the DOM and digestate were examined using FTIR (Fig. 6d). Two noteworthy characteristic

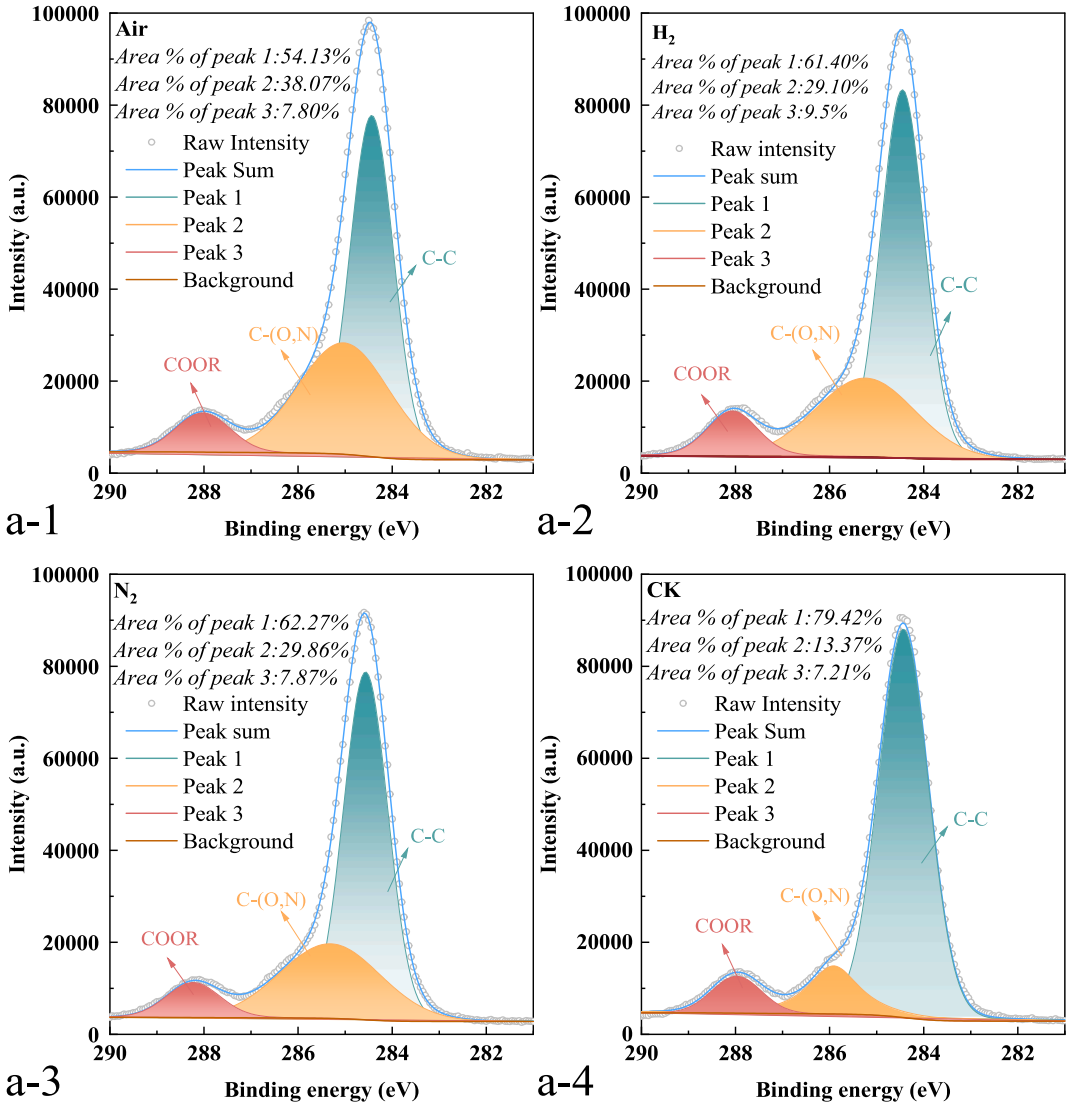


Fig. 6. Effect of NBW on the degradation of cow manure and the nature of soluble substances released: (a) XPS C1s analysis of digested sludge, (b) EEM spectroscopy of DOM, XRD analysis of DOM (c-1) and digested sludge (c-2), FTIR analysis of DOM (d-1) and digested sludge (d-2).

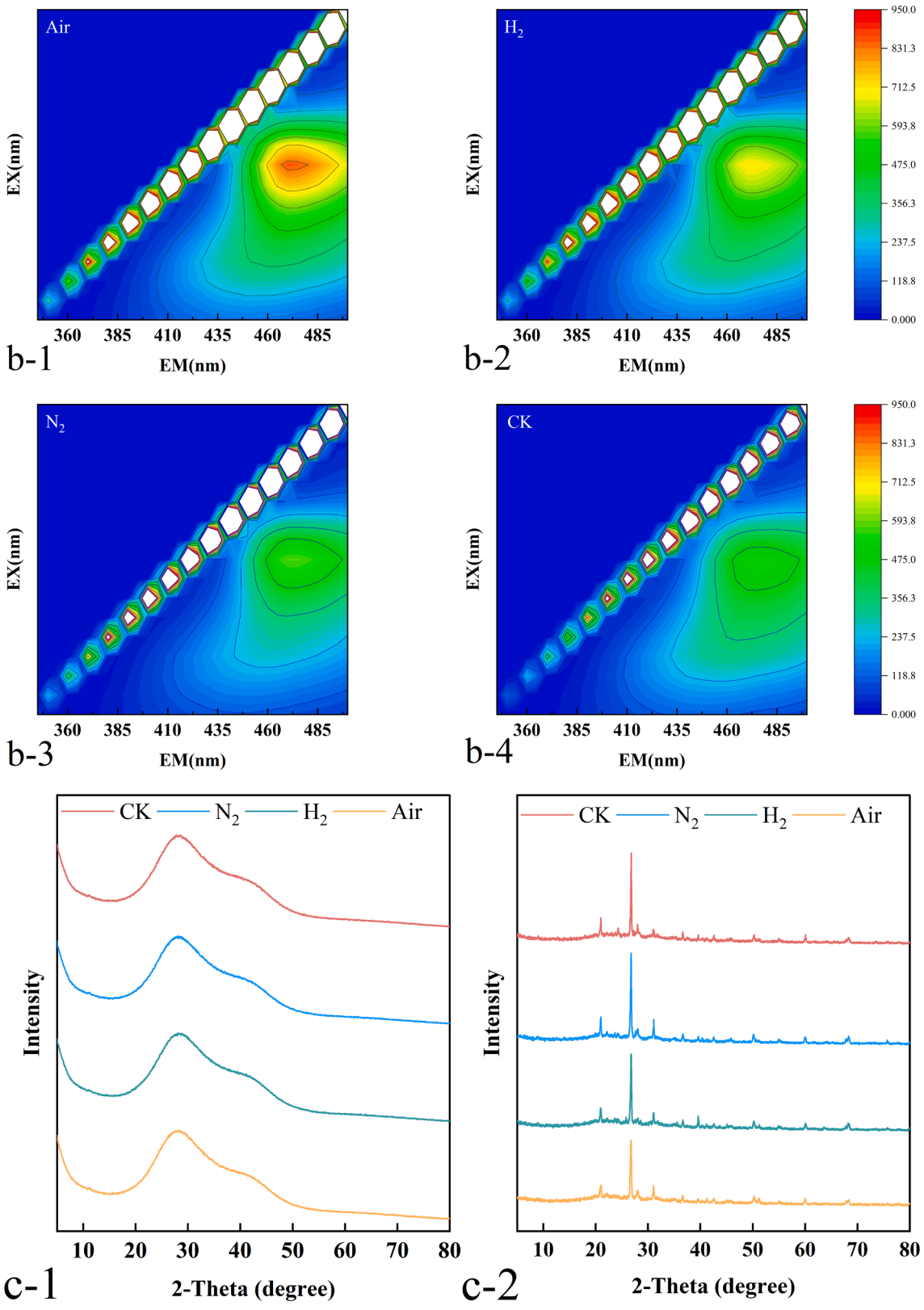


Fig. 6. (*continued*).

peaks related to the stretching vibrations of O-H bonds in C═O, C-N, and hydroxyl functional groups of proteins were observed at 3238 and 1641 cm^{-1}, respectively, in DOM. The FTIR patterns of the digestate presented noteworthy variations in the C-H stretching of C–OH and C–O–C in polysaccharides or carbohydrates at 1038 and 2905 cm^{-1} for CH_2 and CH_3, respectively (Wang et al., 2023d). The experimental group treated with NBW showed improvements compared with the CK experimental group, and the improvement in the experimental group treated with Air was the most notable. The substantial increase in intensity observed in the Air experimental group indicates that the Air-NBW treatment

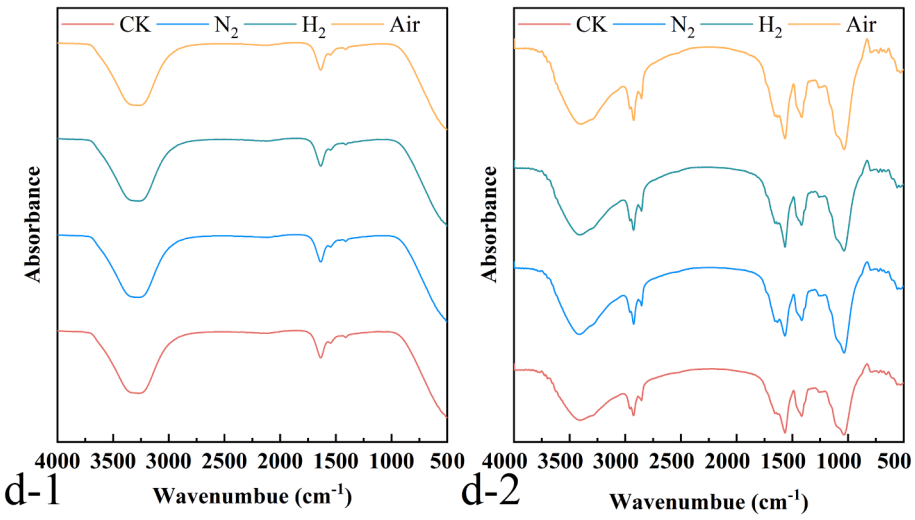


Fig. 6. (*continued*).

facilitates the decomposition of organic materials in cow manure.

3.5. Microbial analysis

To elucidate the mechanism of action of NBW, we evaluated the microbial community in each experimental group during AD (Fig. 7). The highest proportion of Firmicutes was observed in all the experimental groups at the phylum level (Fig. 7a). However, NBW did not have a notable impact on Firmicutes, which is recognized for breaking down complex organic matter, and most CE microorganisms are classified as Firmicutes (Wu et al., 2018). The relative abundances of Actinobacteria in the Air experimental group, H_2 experimental group, N_2 experimental group and CK experimental group were 5.76, 6.26, 6.06 and 5.19 %, respectively. The promotion of MCCAs production by Actinobacteriota is achieved through the release of extracellular enzymes that decompose organic matter macromolecules into VFAs (Zhang et al., 2023). The abundance of Bacteroidota was 7.26 % in the Air experimental group, 6.65 % in the H_2 experimental group, and 4.89 % in the N_2 experimental group. These values were 113.53, 95.59, and 43.82 % higher, respectively, than those in the CK experimental group (3.40 %). The phylum Bacteroides represents a phylum in the CE system comprising organic degrading bacteria, and this phylum plays a vital function in MCCAs production (Wang et al., 2023a). Compared with the CK experimental group (0.91 %), the relative abundances of Halobacterota in the Air, H_2, and N_2 experimental groups were 2.72, 5.93, and 2.13 %, respectively. These values were 198.90, 551.65, and 134.07 % greater than those of the CK experimental group. In AD, the Halobacterota phylum is involved in cellulose degradation (Yang et al., 2023). The better substrate degradation in the NBW experimental group was associated with its increased relative abundance.

At the genus level (Fig. 7b), the relative abundance of *Bacillus* was significantly increased by NBW addition, representing 19.21, 43.15, and 24.95 % of bacterial genera in the air, H_2, and N_2 experimental groups ($p < 0.05$), respectively, and these values were 135, 428, and 20 % greater than that of the CK experimental group (8.18 %). It has been reported that *Bacillus* is involved in the production of MCCAs, and the increase in its relative abundance contributes to MCCAs synthesis (Wu et al., 2022a). *Caproiciproducens* is a typical genus of caproic acid-producing bacteria that produces caproic acid using lactic acid as an electron donor (Tang et al., 2022). *Caproiciproducens* was present in all the experimental groups, with the highest relative abundance of 3.99 % in the Air experimental group, which was slightly greater than the 3.93 % in the CK experimental group. *Lactobacillus* degrades organic matter to produce lactic acid for CE, which is related to MCCAs production (Kurniawan et al., 2023). The relative abundance of bacteria in the air experimental group was 2.48 %, which was greater than that in the CK experimental group (1.78 %). In the AD system, *Syntrophomonas* enables electron transfer between different bacteria (Zhao et al., 2018), and the relative abundance in the Air experimental group was 2.25 %, which was 70.45 % greater than that in the CK experimental group (1.32 %).

Tax4 Fun predicted the metabolic functions of microbial communities following the introduction of various NBW (Fig. S4). The enhancement of carbohydrate metabolism, amino acid metabolism and membrane transport metabolic pathways suggested that the addition of Air-NBW enhanced the degradation of CM and provided more substrates for CE. This analysis further revealed that the relative abundance of ABC transporters (9.30 %) was significantly ($p < 0.05$) greater in the Air experimental group than in the CK experimental group (7.96 %), and an increase in the relative abundance of ABC transporters enhances AD microbial activity and promotes hemicellulose degradation (Zhang et al., 2022a). We also analysed the expression of key enzyme-encoding genes involved in the CE reaction in the Air experimental group and the control group (Fig. S4d). The relative abundance of genes related to the RBO pathway decreased, whereas the relative abundance of genes related to the FAB pathway (6.4.1.2) increased. The abundance in the Air experimental group was 0.3637 %, which was 11.66 % greater than that in the control group. The addition of NBW simultaneously promotes FAB and carbohydrate metabolism, resulting in increased MCCAs production potential (Yu et al., 2023).

3.6. Implications

This study presented a promising approach to improve the production of MCCAs from cow manure AD by adding NBW. The results demonstrated that Air-NBW had a significant impact on increasing MCCAs production, with caproic acid concentration reaching 15.10 g/L, which is 55.03 % greater than that of the CK experimental group. Correlation analysis was performed based on the changes in key physicochemical properties and MCCAs concentrations during the AD process (Fig. 8a). A positive correlation was noted between the caproic acid concentration and the EAC and LDH and a negative correlation was noted between the caproic acid concentration and the lactic acid concentration. Consequently, it is plausible that the higher caproic acid concentration in the Air experimental group was due to the following relationships (Fig. 8b): (1) Air-NBW enhanced the electron transfer capacity of the AD system, accelerating the redox reaction of the CE process; (2) Air-NBW facilitated the decomposition of recalcitrant organic matter in cow manure, leading to an improved substrate utilization rate;

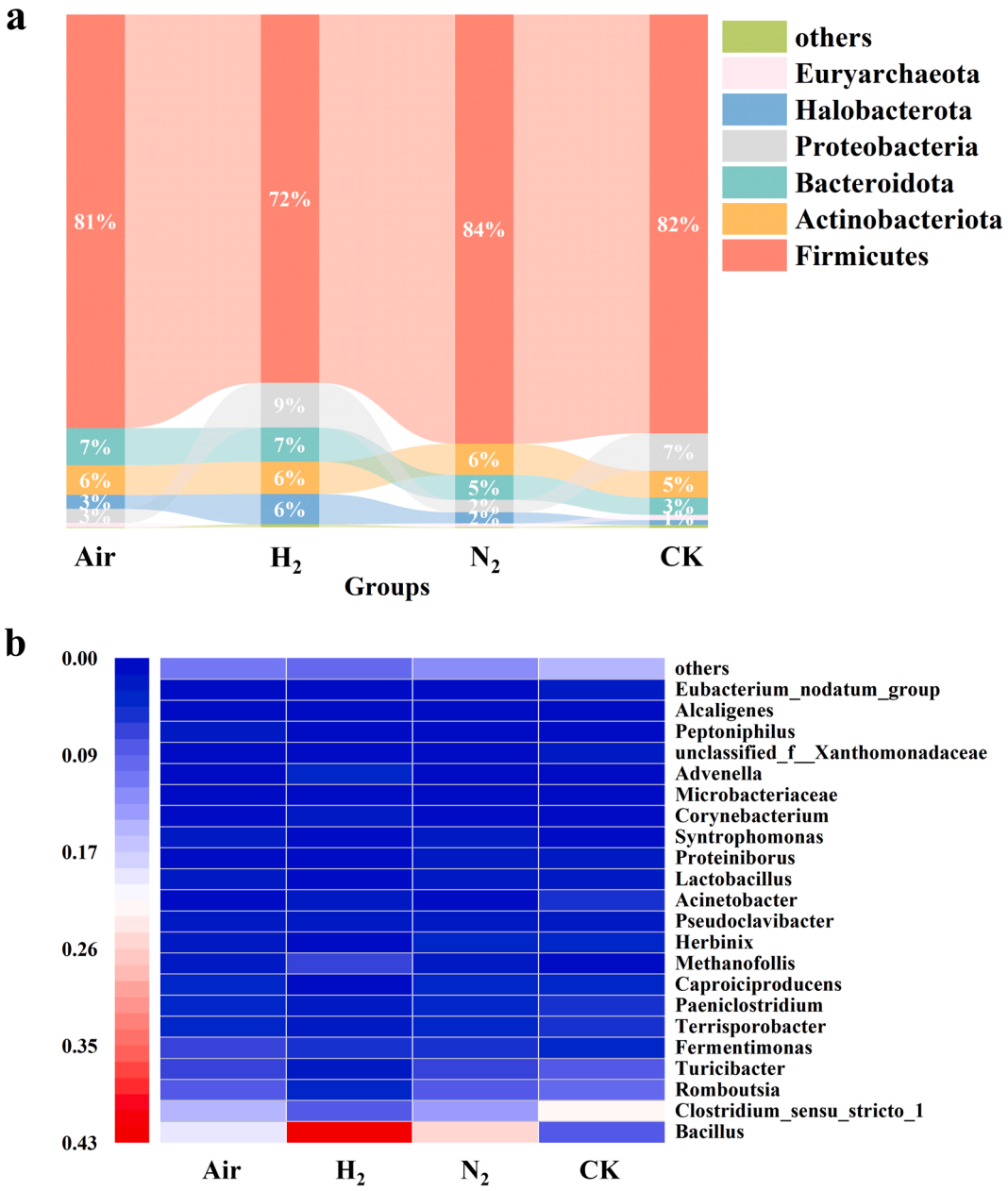


Fig. 7. Relative abundance of bacteria at the phylum level (a) and genus level (b).

(3) Air-NBW decreased the ammonia nitrogen concentration in the AD system, thereby maintaining it in a favorable ammonia-nitrogen environment; (4) Air-NBW boosted the abundance of CE-related *Caproiciproducens* for MCCAs synthesis; (5) the increase in the abundance of the commensal bacterium *Syntrophomonas* and the increase in electron transfer capacity enhanced the efficiency of electron transfer between different bacteria, improving reaction activities; and (6) the enhancement of FAB metabolic pathways, carbohydrate metabolism, and ABC transporters promoted cow manure degradation and increased the production potential of MCCAs.

The introduction of Air-NBW altered the microbial community structure, increasing the relative abundance of microorganisms associated with MCCAs production and enhancing electron transfer among different bacteria, thereby enhancing the carbohydrate metabolism pathway. Microorganisms employ exogenous and endogenous mechanisms for survival in diverse environments, perceiving changes in chemical gradients and evolving towards conditions conducive to microbial growth through a phenomenon known as chemotaxis (Fang et al., 2024; Miller et al., 2009). NBW directly disrupts the structure of recalcitrant organic compounds such as lignocellulose, facilitating degradation by hydrolytic bacteria. Additionally, prior to rupture, NBW facilitates electron transfer, enhancing interspecies electron transfer among microorganisms and the activity of electron transfer systems. These effects of NBW induce changes in the chemical environment of AD systems. The chemotaxis of anaerobic bacteria prompts functional microorganisms to demonstrate an enhanced adaptive capacity for substrate degradation and MCCAs production.

4. Conclusion

In this study, we evaluated the effect of different types (air, H_2, and N_2) of NBW on MCCAs production from cow manure AD. Air-NBW had the best effect, with caproic acid concentration reaching 15.10 g/L, which was 55.03 % greater than that of the CK experimental group. Air-NBW enhanced electron transfer and substrate degradation and reduced ammonia nitrogen. This process stimulates changes in the structure of

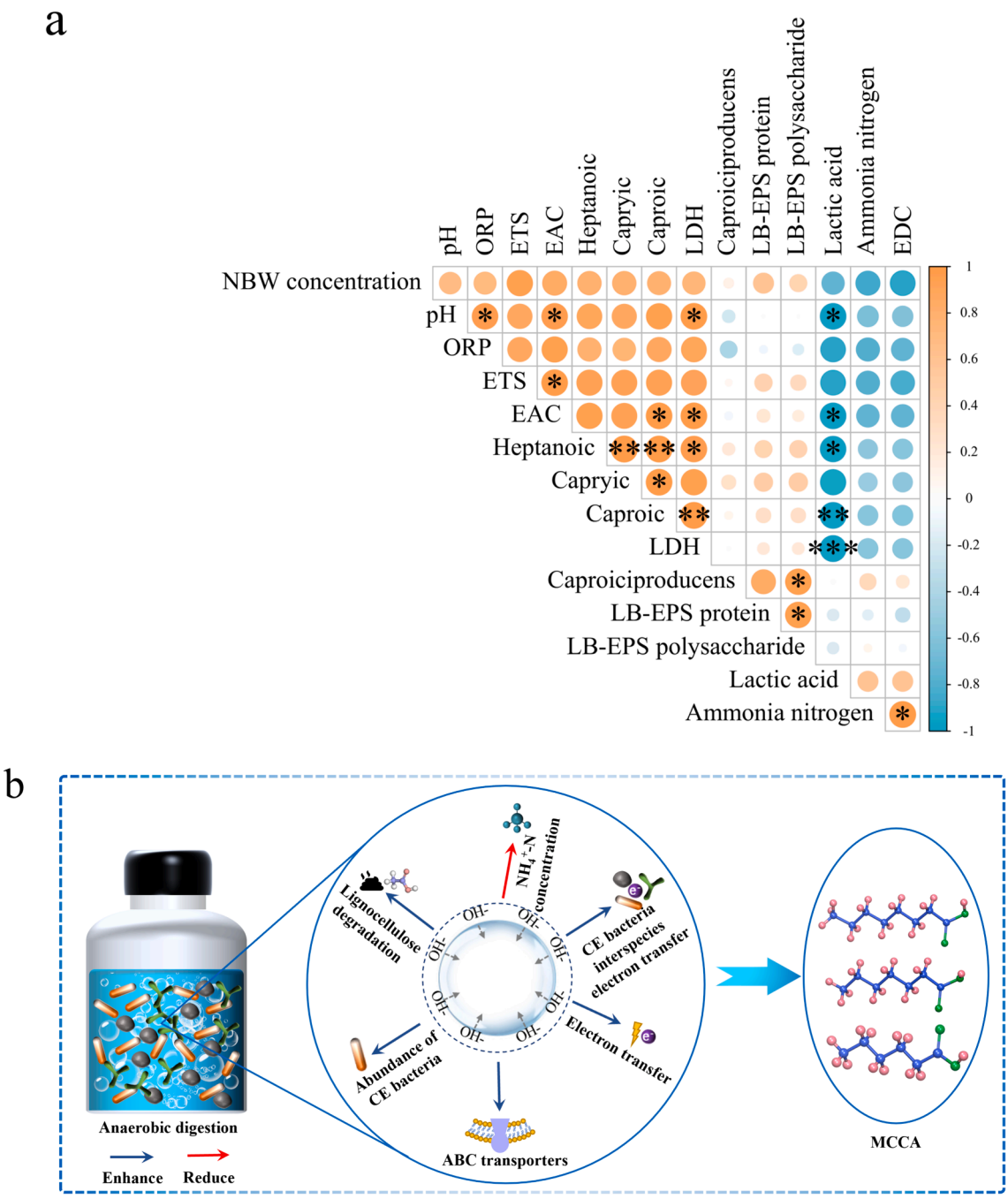


Fig. 8. (a) Correlation between basic indicators and MCCAs concentrations under different treatments of MCCAs fermentation process. Both the depth and size of the circle colors indicate the strength of the correlation (*, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$); (b) Intensification pathway of NBW to MCCAs production from cow manure AD.

the microbial community and enhances carbohydrate metabolic pathways, which subsequently promote substrate degradation. The relative abundance of *Bacillus* and *Caproiciproducens* involved in CE increased, the relative abundance of *Syntrophomonas* involved in electron transfer increased, and the microbial community was optimized. The FAB cycle pathway and carbohydrate metabolism pathway of the CE process were strengthened. Overall, this technology has ample potential for producing MCCAs from the AD of organic waste.

CRediT authorship contribution statement

Yang Liu: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Xiaomei Ye:** Writing – review & editing, Supervision, Conceptualization. **Kequan Chen:** Supervision, Conceptualization. **Xiayuan Wu:** Methodology, Formal analysis, Data curation. **Lihua Jiao:** Validation, Resources. **Hongyu Zhang:** Validation, Methodology. **Fei Zhu:** Resources, Data curation. **Yonglan Xi Ye:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.wasman.2024.05.029.

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